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Methylation Level & Entropy Simulation

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Description	This document outlines a computational pipeline designed to analyze and visualize methylation entropy from Whole-Genome Bisulfite Sequencing (WGBS) data. The primary goal is to distinguish between changes in methylation entropy that naturally arise from varying methylation levels, and those that might suggest active biological regulation driving methylation heterogeneity.

1. Overview & Research Question

Research Question:

Are observed changes in DNA methylation entropy primarily a consequence of changes in overall methylation levels (where entropy naturally increases as methylation approaches 50%), or do they reflect active biological mechanisms that maintain a specific high (or low) entropy state independent of the mean methylation level? This pipeline aims to address this by comparing observed entropy to theoretical minimum and stochastic (random) entropy baselines.

2. Pipeline Components

This pipeline consists of 3 main parts:

A. Pre-computation of Minimum Entropy Lookup Table

▼ Code:

`create_lookup.py`

create_lookup.py

▼ Purpose:

This Python script systematically calculates the *absolute minimum possible methylation entropy* for all possible total numbers of methylated CpGs, given a fixed number of reads (16) and CpGs (4) per read. This serves as a theoretical lower bound for methylation entropy

▼ How it works:

It exhaustively searches through combinations of read types (e.g., reads with 0, 1, 2, 3, or 4 methylated CpGs) to find the configuration that minimizes entropy for a specific overall methylation count. The results are stored in a JSON file for quick retrieval.

▼ Key output:

`minimum_entropy_lookup.json` (a dictionary mapping methylation levels to their minimum possible entropy values)

minimum_entropy_lookup.json

#comment: I used the 16-read, 4-CpG-per-read setting for this calculation to minimize computational load. However, for the later simulations, I used 128 reads instead—so this difference should be noted

B. Main Entropy Simulation and Calculation

▼ Code: `entropy_simulation_raw_multi_final.py`

entropy_simulation_raw_multi_final.py

▼ Purpose:

This Python script takes raw WGBS entropy data as input (generated from Sara's RLM and selected the methylation and entropy column) and enriches it by calculating the corresponding minimum (allelic) and stochastic (random) entropy ranges for each genomic locus.

▼ How it works:

1. Loads observed methylation entropy data (RLM result containing observed entropy and methylation level)
2. Uses the pre-computed `minimum_entropy_lookup.json` to determine the theoretical minimum entropy for each observed methylation level
3. Performs Monte Carlo simulations to generate a *stochastic entropy range*. For each locus's observed methylation level, it simulates `n_sims` (default 1000) scenarios where methylated CpGs are randomly distributed across `n_reads` (default 128) DNA molecules. This yields a mean, 5th percentile (min), and 95th percentile (max) for entropy expected by random chance.
4. Combines these calculated values with original data
5. Utilizes `multiprocessing` to efficiently process large datasets.

▼ Usage (Example for HPC via SLURM):

See `simulation_hpc_raw_mef.sh` below

▼ Key output:

Tab-separated files containing: `chrom`, `start`, `end`, `observed_entropy`, `methylation`, `allelic_entropy`, `stochastic_mean`, `stochastic_min`, `stochastic_max`



C. HPC Job Submission Script

▼ Code:

simulation_hpc_raw_mef.sh

D. Visualization of Simulation Results (Metaplots)

▼ Code:

example_mef_metaplots.Rmd

▼ Purpose:

This R Markdown script visualizes the output from `entropy_simulation_raw_multi_final.py` to create metaplots. These plots show the average profiles of observed, stochastic, and allelic entropy, as well as methylation levels, centered around specific genomic features (like DMVs) across different cell passages. This allows for direct visual comparison to interpret the nature of methylation heterogeneity

▼ How it works:

1. **Loads Data:** Reads your pre-defined DMV (Differentially Methylated Regions) and CGI (CpG Island) genomic coordinates, and the processed entropy data from the Python output files
2. **Creates Normalized Matrices:** Uses `EnrichedHeatmap::normalizeToMatrix` to calculate average entropy/methylation values in bins surrounding the target genomic features (DMVs in this case). This creates matrices where each row is a feature and columns are spatial bins
3. **Prepares Plot Data:** Summarizes these matrices by calculating the mean, 25th percentile, and 75th percentile for each spatial bin, creating data suitable for `ggplot2`
4. **Generates Plots:** Uses `ggplot2` to create metaplots with ribbons for percentile ranges and smoothed lines for means. It plots observed entropy, stochastic entropy (5-95th percentile range from simulation output, then summarized to 25-75th percentile across features), allelic (minimum) entropy, and methylation levels.

▼ Key Output:

PDF files containing the metaplots

▼ Considerations:

1. `extend` **and** `w` **parameters:** These control the genomic window size and bin resolution around target regions. Adjust them as needed for specific biological question.
2. **Percentile interpretation:** Be clear about what the ribbons represent. In this R code, we're plotting the 25th and 75th percentiles *across the DMVs* for each entropy type. For "stochastic", this means it's the 25th/75th percentile of the `stochastic_min` and `stochastic_max` values (which were already the 5th/95th percentiles from the original simulations).

3. Interpretation of Results (General Guide)

When examining the metaplots, focus on the relationships between the four lines/ribbons:

- **Observed (seagreen):** Your actual data.
- **Stochastic (royalblue):** The entropy expected if methylation patterns were purely random for a given methylation level.
- **Allelic-specific (red):** The theoretical minimum entropy.
- **Methylation (black):** The average methylation level

What to look for:

- **Observed within Stochastic Range:** If the observed entropy line falls primarily within the stochastic ribbon, it suggests that the observed methylation heterogeneity at those regions can be largely explained by random chance given the mean methylation level.
- **Observed significantly *above* Stochastic Range:** If the observed entropy is consistently higher than the stochastic range, it indicates that cells are actively maintaining a more heterogeneous methylation state than would occur randomly. This points towards biological regulation promoting epigenetic variability.
- **Observed close to Allelic-specific:** If the observed entropy is very close to the allelic minimum, it suggests a highly uniform and potentially tightly regulated methylation pattern, where most molecules within the population are methylated in a very similar way.

4. Real Data analysis & Results